RULE-BASED LIBRARY CHATBOT

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1. Introduction

A rule-based chatbot is a computer program that responds to user questions using predefined rules. This project uses Python and if-else conditions to identify keywords in user messages and provide suitable answers.

2. Selected Topic

Library Chatbot

The chatbot provides basic information about library timings, membership, borrowing books, returning books, late fines, and book searches.

3. Objectives

- To develop a simple chatbot using Python.
- To understand if-else conditional statements.
- To accept user input and display appropriate responses.
- To handle unknown questions with a default response.

4. Sample Chatbox Conversations

Conversation 1: Greeting

User: Hello

Bot: Hello! Welcome to the library. How can I help you?

Conversation 2: Library Timings

User: What are the library opening timings?

Bot: The sample library timing is 9:00 AM to 6:00 PM, Monday to Saturday.

Conversation 3: Membership

User: How can I get library membership?

Bot: Please fill out the membership form and show your valid ID.

Conversation 4: Borrowing Books

User: How many books can I borrow?

Bot: Members can borrow up to 3 books at a time for 14 days. This is a sample library policy.

Conversation 5: Returning Books

User: How do I return a book?

Bot: Please return borrowed books to the library desk before or on the due date.

Conversation 6: Late Fine

User: What is the fine for returning a book late?

Bot: A sample late-return fine is ₹5 per book per day. Please confirm the actual fee with library staff.

Conversation 7: Searching for a Book

User: How can I find a Python book?

Bot: Search the library catalogue by book title, author, or subject, or ask the librarian for help.

Conversation 8: Unknown Question

User: Can you book a taxi for me?

Bot: Sorry, I don't understand that question. Please ask about library timings, membership, borrowing, returns, fines, or book searches.

Conversation 9: Ending the Chat

User: Bye

Bot: Goodbye! Thank you for visiting the library chatbot.

5. Working Principle

The chatbot accepts a question from the user and converts the input into lowercase. It checks the message against predefined conditions using if-elif-else statements. If a condition matches, the chatbot displays the corresponding answer. Otherwise, it displays a default response. The conversation ends when the user types “bye”, “exit”, or “quit”.

6. Example Output

WELCOME TO THE LIBRARY CHATBOT
Type 'bye' to end the conversation.

You: Hi
Bot: Hello! Welcome to the library. How can I help you?

You: How many books can I borrow?
Bot: Members can borrow up to 3 books at a time for 14 days (sample library policy).

You: bye
Bot: Goodbye! Thank you for visiting the library chatbot.

7. Conclusion

This project demonstrates how a simple rule-based chatbot can be created using Python. It answers common library questions using predefined conditions and provides a default response when it cannot understand a question.